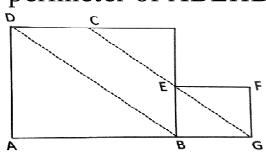
AMTI (NMTC) - 2012

KAPREKAR CONTEST - SUB-JUNIOR LEVEL

PART - A

1. p is an odd prime. Then $\left(\frac{p+1}{2}\right)^2 - \left(\frac{p-1}{2}\right)^2$ is
   (A) a fraction less than p. (B) a fraction greater than p.
   (C) a natural number not equal to p. (D) a natural number equal to p.
2. PQRS is a parallelogram. MP and NP divide ∠SPQ into three equal parts (∠MPQ > ∠NPQ) and MQ and NQ divide ∠RQP into 3 equal parts (∠MQP > ∠NQP). If k(∠PNQ) = (∠PMQ) then k =
   (A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) $\frac{1}{3}$
3. In a foot ball league, a particular team played 50 games in a season. The team never lost three games consecutively and never won four games consecutively, in that season. If N is the number of games the team won in that season, then N satisfies
   (A) $25 \leq N \leq 30$ (B) $25 \leq N \leq 36$ (C) $16 \leq N \leq 38$ (D) $16 \leq N \leq 30$
4. a, b, c, d, e are five integers such that a + b = b + c = c + d = d + e = 2012, a + b + c + d + e = 5024. Then the value of (d – a) is
   (A) 8 (B) 12 (C) 10 (D) 4
5. The last two digits of 3^{2012}, when represented in decimal notation, will be
   (A) 81 (B) 01 (C) 41 (D) 21
6. There exists positive integers x, y such that both the expressions (3x + 2y) and (4x – 3y) are exactly divisible by
   (A) 11 (B) 7 (C) 23 (D) 17
7. The years of 20th century and 21st century are of 4 digits. The number of years which are divisible by the product of the four digits of the year is
   (A) 7 (B) 8 (C) 9 (D) none of these
8. If $x+\frac{1}{x}=-1$, then the value of $x^3-\frac{1}{x^3}$ is
   (A) 0 (B) 1 (C) –1 (D) 2
9. The number of natural numbers a (< 100) such that $(a^3 - a^2)$ is a square of a natural number is
   (A) 7 (B) 8 (C) 9 (D) 10
10. Let n be the smallest nonprime integer greater than 1 with no prime factor less than 10. Then
    (A) $100 < n \leq 100$ (B) $110 \leq n \leq 120$
    (C) $120 < n \leq 130$ (D) $130 < n \leq 140$
11. A point P inside a rectangle ABCD is joined to the angular points. Then
    (A) Sum of the areas of two of the triangles so formed is equal to the sum of the other two.
    (B) The sum of the areas of the triangles so formed is a whole number whatever may be the dimensions of the rectangle.
    (C) The sum of the areas of a pair of opposite triangles is greater than half the area of the rectangle.
    (D) None of these
12. In the adjoining figure ABCD and BGFE are rhombuses. AB = 10 cm, GF = 3 cm. GE meets DC at H. ∠A = 60^0. The perimeter of ABEHD is (in cm)



    (A) 47 (B) 40 (C) 39 (D) 33

13. If $3a + 1 = 2b - 1 = 5c + 3 = 7d + 1 = 15$, then the value of $(3a - b + 5c - 9d)$ is
(A) 0 (B) 1 (C) 2 (D) 3

14. The perimeter of an isosceles right angled triangle is 2012. Its area is
(A) $2012(3 - \sqrt{2})$ (B) $(1006)^2(3 - \sqrt{2})$
(C) $(2012)^2$ (D) $(1006)^2$

15. Five two digit numbers (none of the digits is zero) add up to 100. If each digit is replaced by its 9 complement, then the sum of these five new numbers is
(A) 295 (B) 195 (C) 380 (D) 395

PART - B

1. The sum of two natural numbers is 484. Their HCF is 11. The number of such possible pairs is _______.

2. ABCD is a trapezium with AB and CD parallel. If AB = 16 cm, BC = 17 cm, CD = 8 cm, DA = 15 cm then the area of the trapezium (in cm^2) is ______.

3. The number of multiples of 9 less than 2012 and having sum of the digits as 18 is ____.

4. AB, CD, EF, GH and IJ are five 2-digit numbers such that each letter stands for a different digit. The largest possible sum of these five numbers is _______.

5. Two consecutive natural numbers are respectively by 4 and 7. The sum of the their respective quotient is 8. Then the sum between the numbers is ______.

6. p is the difference between a real number and its reciprocal. q is the difference between the square of the same real number and the square of the reciprocal. Then the value of $p^4 + q^2 + 4p^2$ is _____.

7. A square is inscribed in another square each of whose four vertices lies on each side of the square. The area of the smaller square is $\frac{25}{49}$ times the area of the bigger one. Then the ratio with which each vertex of the smaller square divides the side of the bigger square is ______.

8. a, b, c are three positive numbers. The second number is greater than the first by the amount the third number is greater than the second. The product of the two smaller numbers is 85 and that of the two bigger numbers is 115. Then the value of $(2012a - 1006c)$ is ______.

9. If $x = \frac{y}{y+1}$ and $y = \frac{a-2}{2}$ the value of $x(y+2) + \frac{x}{y} + \frac{y}{x}$ when a = 2012 is ______.

10. PSR is an isosceles triangle in which PS = PR. SP is produced to O such that PO = SP. Then $\angle$SRO is equal to _______.

11. a, b, c, d are five integers such that $a + b = b + c = c + d = d + e = 2012$ and $a + b + c + d + e = 5024$. Then the value of $(d - a)$ = ______.

12. A class contains three girls and four boys. Every Saturday, five students go on a picnic, a different group is sent each week. During the picnic, each person (boy or girl) is given a Cake by the accompanying teacher. After all possible groups of five have gone once; the total number of cakes received by the girls during the picnic is _______.

13. CAB is an angle whose measure is 70^0. ACFG and ABDE are squares drawn outside the angle. The diagonal FA meets BE at H. Then the measure of the angle EAH is ______.

14. The number of prime numbers p for which p + 2 and are also prime number is _____.

15. The number of integers pairs (m, n) which satisfied $m(n^2 + 1) = 48$ is ______.
